Physica Scripta

INCLUSIVE PUBLISHING TRUSTED SCIENCE

PURPOSE-LED PUBLISHING™

The Nuclear Potential-energy Surface with Inclusion of Axial Asymmetry

To cite this article: S E Larsson 1973 *Phys. Scr.* **8** 17

View the article online for updates and enhancements.

You may also like

- The Optical/Infrared Astronomical Quality of High Atacama Sites. I. Preliminary Results of Optical Seeing
Riccardo Giovanelli, Jeremy Darling, Marc Sarazin et al.
- Synthesis and study of properties of superheavy atoms. Factory of Superheavy Elements
Yuri Ts. Oganessian and Sergey N. Dmitriev
- (Invited) Achieving Vertical Trench-Gate GaN MOSFETs via Process Optimization
David I Shahin, Travis J Anderson and Aristos Christou

This content was downloaded from IP address 131.112.127.17 on 30/10/2025 at 05:30

Physica Scripta. Vol. 8, 17–31, 1973

The Nuclear Potential-energy Surface with Inclusion of Axial Asymmetry

S. E. Larsson

Department of Mathematical Physics Lund, Institute of Technology, Lund, Sweden

Received January 19, 1973

Abstract

The nuclear potential-energy surface with inclusion of axial asymmetry. S. E. Larsson (Department of Mathematical Physics, Lund Institute of Technology, Lund, Sweden).
Physica Scripta (Sweden) 8, 17–31, 1973.

The nuclear potential-energy surface has been calculated on the basis of the modified oscillator model, including the degree of freedom of axial asymmetry. The regions investigated are the prolate-oblate transitional region around Os-Pt, the actinide region and the region of superheavy elements. The theoretical results indicate that the first barrier of the two-peak barrier in the actinides is reduced due to axial asymmetry, which leads to an improved agreement with empirical data. There also appears to be a modest reduction of the barrier for superheavy elements.

1. Introduction

On the path to fission, nuclei pass through a series of widely varying shapes. Until recently most calculations of the nuclear potential energy surface have been limited to shapes that are usually both axially symmetric and reflection asymmetric [1].

A priori, there is no reason to study only axially symmetric shapes. In an early paper dealing with the fission process, Pashkevich [2] considered axial asymmetric shapes. These calculations have later been extended by Schultheiss et al. [3, 4], Götz et al. [5] and Larsson et al. [6]. The formally complete description of the total energy surface in terms of the ellipsoidal shape coordinates β and γ was introduced in the early papers by A. Bohr [7] and A. Bohr and B. Mottelson [8], who also developed a general dynamical formulation.

Spectroscopic data so far available have been taken to indicate the strong preference in nature for relatively pure prolate spheroidal shapes in the established regions of nuclear deformations. This is true at least for larger A-values.

However, the sophisticated dynamical calculations by Kumar and Baranger [9–11] based on the pairing plus quadrupole force model, where the dynamical mass parameters are calculated microscopically, indicate that at the end of the rare-earth region there may be a transitional region, involving isotopes of W, Os and Pt with A close to 190, containing nuclei with triaxial equilibrium shapes well into the γ-plane in addition to pure oblate ones.

There may in fact be other analogous γ-transitional regions near Ba or Te as discussed e.g. in the work by A. Johnson and R. Foucher [12, 13].

2. The nuclear potential field

The present calculations are based on the modified oscillator model, where it is assumed that the interaction of one nucleon with the nuclear field can be described by an one-body Hamiltonian:

$$H = T + \frac{1}{2}\hbar\omega_0\varrho^2\left\{1 - \frac{2}{3}\varepsilon\sqrt{\frac{4\pi}{5}}\left[\cos\gamma Y_{20} - \frac{\sin\gamma}{\sqrt{2}}(Y_{22} + Y_{2-2})\right]\right\}$$
$$- \varkappa\hbar\omega_0^0\{2\boldsymbol{l}_t\boldsymbol{s} + \mu(\boldsymbol{l}_t^2 - \langle\boldsymbol{l}_t^2\rangle_N)\}$$
$$= \frac{1}{2}\hbar\omega_0(-\Delta + \varrho^2) - \hbar\omega_0\frac{2}{3}\varepsilon\sqrt{\frac{4\pi}{5}}\varrho^2\left[\cos\gamma Y_{20} - \frac{\sin\gamma}{\sqrt{2}}(Y_{22} + Y_{2-2})\right]$$
$$- \varkappa\hbar\omega_0^0\{2\boldsymbol{l}_t\boldsymbol{s} + \mu(\boldsymbol{l}_t^2 - \langle\boldsymbol{l}_t^2\rangle_N)\} = H_0 + H_\varepsilon + H_{\text{corr}} \tag{1}$$

corresponding to the three frequencies

$$\omega_x = \omega_0(\varepsilon, \gamma)\left[1 - \tfrac{2}{3}\varepsilon\cos\left(\gamma + \frac{2\pi}{3}\right)\right]$$
$$\omega_y = \omega_0(\varepsilon, \gamma)\left[1 - \tfrac{2}{3}\varepsilon\cos\left(\gamma - \frac{2\pi}{3}\right)\right] \tag{2}$$
$$\omega_z = \omega_0(\varepsilon, \gamma)\,[1 - \tfrac{2}{3}\varepsilon\cos\gamma]$$

where ε and γ specify the degree of deformation and the departure from axial symmetry, respectively. This type of nonisotropic oscillator was studied in detail e.g. in ref. [14].

In the expression for the potential we have used triaxially stretched coordinates [15]

$$\xi = x\sqrt{\frac{M\omega_x}{\hbar}}\ \text{etc.} \tag{3}$$

and

$$\varrho^2 = \xi^2 + \eta^2 + \zeta^2 \tag{4}$$

As shown in ref. [15] this choice of coordinates has the advantage of transforming away coupling terms of $\varrho^2 Y_{20}$ and $\varrho^2(Y_{22} + Y_{2-2})$ between oscillator shells N and $N \pm 2$ (see the next section).

As seen above the potential also contains the usual type of spin-orbit term and the special l_t^2-term [16], the latter having the effect of making the potential more flat-bottomed. The operator $\boldsymbol{l}_t$ is the orbital angular momentum constructed in terms of the stretched coordinates and the corresponding canonically conjugate momentum operators.

In this formulation we can easily include other multipoles, i.e. terms proportional to $\varepsilon_4\,\varrho^2 P_4$, $\varepsilon_6\,\varrho^2 P_6$, $\varepsilon_{33}\,\varrho^2 Y_{33}$ etc.

We have usually limited the present calculations to not including P_4-deformations. The effect of deformations with $\varepsilon_4 \neq 0$ is to lower the energy on the prolate side more than on the oblate side. The reason not to include them here is the fact, that they uniquely favour the prolate side as the most relevant degree of freedom there. On the oblate side, on the other hand, there should rather be a term proportional to $\varepsilon_{44}\,\varrho^2 Y_{44}$, which we have not been able to include in the present calculations [17].

3. Hamiltonian matrix elements

The three different oscillator frequencies can also be written in terms of a new set of deformation parameters ε_2 and ε_{22}:

$$\begin{aligned}\omega_x &= \omega_0(\varepsilon_2, \varepsilon_{22})\,[1 + \tfrac{1}{3}\varepsilon_2 + \tfrac{1}{2}\varepsilon_{22}]\\ \omega_y &= \omega_0(\varepsilon_2, \varepsilon_{22})\,[1 + \tfrac{1}{3}\varepsilon_2 - \tfrac{1}{2}\varepsilon_{22}]\\ \omega_z &= \omega_0(\varepsilon_2, \varepsilon_{22})\,[1 - \tfrac{2}{3}\varepsilon_2]\end{aligned} \tag{5}$$

with the definitions:

$$\varepsilon_2 = \varepsilon \cos\gamma$$

and

$$\varepsilon_{22} = \varepsilon \frac{\sin\gamma}{\sin\frac{\pi}{3}} \tag{6}$$

The Hamiltonian in terms of ε_2 and ε_{22} then takes the form:

$$\begin{aligned}H = {} & \frac{1}{2}\hbar\omega_0\{-\Delta + \varrho^2\} + \frac{1}{6}\varepsilon_2\hbar\omega_0\left\{\left(-\frac{\partial^2}{\partial\xi^2} + \xi^2\right) + \left(-\frac{\partial^2}{\partial\eta^2} + \eta^2\right)\right.\\ & \left. - 2\left(-\frac{\partial^2}{\partial\zeta^2} + \zeta^2\right)\right\} + \frac{1}{4}\varepsilon_{22}\hbar\omega_0\left\{\left(-\frac{\partial^2}{\partial\xi^2} + \xi^2\right) - \left(-\frac{\partial^2}{\partial\eta^2} + \eta^2\right)\right\}\\ & - \varkappa\hbar^0_{\omega_0}\{2\boldsymbol{l}_t\boldsymbol{s} + \mu(\boldsymbol{l}_t^2 - \langle\boldsymbol{l}_t^2\rangle_N\} = H_0 + H_2 + H_{22} + H_{\mathrm{corr}}\end{aligned} \tag{7}$$

where ξ, η, ζ are thus the triaxially stretched coordinates. It has been shown by Nilsson [15] that the term H_2 is diagonal in N. In the same manner one can show, that the term H_{22} also is diagonal in N, utilizing the commutator relations given in Appendix I. Thus, this term has the same matrix elements as

$$\delta_{N'N}\,\varepsilon_{22}\hbar\omega_0\sqrt{\frac{2\pi}{15}}\varrho^2\{Y_{22}(\theta_t, \phi_t) + Y_{2-2}(\theta_t, \phi_t)\} \tag{6}$$

where the relation between the angles θ_t, ϕ_t in stretched coordinates and the ordinary angles in xyz-space are given by

$$\cos\theta_t = \left[\frac{1 - \frac{2}{3}\varepsilon\cos\gamma}{1 + \frac{1}{3}\varepsilon\cos\gamma - \varepsilon\cos\gamma\cos^2\theta + \frac{1}{\sqrt{3}}\varepsilon\sin\gamma\sin^2\theta\cos 2\phi}\right]^{1/2} \times\cos\theta \tag{8}$$

and

$$\tan\phi_t = \left(\frac{\omega_y}{\omega_x}\right)^{1/2}\tan\phi \tag{9}$$

It is easy to verify that neither l, j nor Ω are conserved, while obviously parity is conserved.

The basis functions are chosen as the eigenstates of the isotropic harmonic oscillator

$$|Nl\Lambda\Sigma\rangle \sim e^{-\varrho^2/2}F\left(-\frac{N-l}{2}, l + \frac{3}{2}; \varrho^2\right)\varrho^l Y_{l\Lambda}\chi^{\Sigma}_{1/2}(\sigma) \tag{10}$$

where F is a confluent hypergeometric function. This is the most commonly used set of basis states even for one-body potentials that are very different from the oscillator.

The shell model states are thus built from independent single-particle states in the common potential. The correlation between the particles in the wave function is established through the action of some type of residual forces.

The term proportional to $\varrho^2(Y_{22} + Y_{2-2})$ in the potential gives rise to a coupling between states $\Omega \to \Omega \pm 2$.

The necessary and sufficient range of the γ-coordinate can be determined from a consideration of symmetry operations on the ellipsoidal shape. Invariance under rotation an angle $\pi/2$ about the z-axis, invariance under rotation and angle π about the y-axis and invariance under relabelling of the coordinate axes yields the range of γ as

$$0° \leqslant \gamma \leqslant 60°$$

Furthermore the requirement $\omega_x \geqslant \omega_y \geqslant \omega_z$, i.e. the corresponding semi-axes bearing the relations $a \leqslant b \leqslant c$, will require ε to be positive. Thus $\gamma = 60°$ corresponds to pure oblate shapes.

4. The volume conservation condition

In the microscopic calculation the saturation properties of nuclear matter can be taken roughly into account in terms of a volume conservation condition. Since the nuclear field is a shortrange field, we can approximately fulfill this condition by conserving the nuclear volume enclosed by the equipotential surfaces under any deformation. This can be done for all equipotential surfaces simultaneously if we ignore the $\boldsymbol{ls}$- and $\boldsymbol{l}^2$-dependent terms.

Thus, the conservation of volume corresponds simply to a scaling of ω_0:

$$\begin{aligned}\left(\frac{\omega_0}{\overset{0}{\omega_0}}\right)^3 = {} & \frac{1}{4\pi}\frac{1}{\sqrt{\left(1 + \frac{1}{3}\varepsilon_2 + \frac{1}{2}\varepsilon_{22}\right)\left(1 + \frac{1}{3}\varepsilon_2 - \frac{1}{2}\varepsilon_{22}\right)\left(1 - \frac{2}{3}\varepsilon_2\right)}}\\ & \times\int_0^{\pi}\int_0^{2\pi}\frac{\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi}{\left[1 - \frac{2}{3}\varepsilon_2 P_2 + \varepsilon_{22}\sqrt{\frac{2\pi}{15}}(Y_{22} + Y_{2-2}) + 2\varepsilon_4 P_4\right]^{3/2}}\end{aligned} \tag{11}$$

In the case $\varepsilon_4 = 0$, to which we have restricted ourselves, the integral can be evaluated exactly giving the result:

$$\left(\frac{\omega_0}{\overset{0}{\omega_0}}\right)^3 = \frac{1}{\left(1 + \frac{1}{3}\varepsilon_2 + \frac{1}{2}\varepsilon_{22}\right)\left(1 + \frac{1}{3}\varepsilon_2 - \frac{1}{2}\varepsilon_{22}\right)\left(1 - \frac{2}{3}\varepsilon_2\right)} \tag{12}$$

If $\varepsilon_4 \neq 0$ the integral has to be evaluated numerically.

5. The Coulomb and surface energies

In the Swiatecki-Strutinsky shell correction method of calculation of the nuclear potential-energy surface, the Coulomb and surface energies represent the macroscopic part, or the "background" part, of the potential energy.

The general expression for the Coulomb energy is

$$E_{\mathrm{coul}} = \frac{1}{2}\sum_{i\neq j}\sum\left\langle\psi(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_Z)\left|\frac{e^2}{r_{ij}}\right|\psi(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_Z)\right\rangle \tag{13}$$

where $|\psi(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_Z)\rangle$ is the Slater determinant built from single particle states in the potential. If we neglect the exchange energy under the assumption that it does not depend on distor-

Table I. *Coulomb energy as a function of ε and $\gamma(\varepsilon_4 = 0)$*

ε	γ 0°	10°	20°	30°	40°	50°	60°
0.0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
0.1	−0.00092	−0.00092	−0.00091	−0.00089	−0.00087	−0.00086	−0.00086
0.2	−0.00384	−0.00381	−0.00371	−0.00358	−0.00345	−0.00336	−0.00332
0.3	−0.00903	−0.00891	−0.00858	−0.00814	−0.00769	−0.00736	−0.00724
0.4	−0.01682	−0.01653	−0.01574	−0.01466	−0.01358	−0.01279	−0.01251
0.5	−0.02764	−0.02706	−0.02547	−0.02331	−0.02166	−0.01959	−0.01901
0.6	−0.04202	−0.04099	−0.03816	−0.03432	−0.03049	−0.02771	−0.02669
0.7	−0.06070	−0.05898	−0.05431	−0.04797	−0.04170	−0.03713	−0.03547
0.8	−0.08462	−0.08192	−0.07459	−0.06470	−0.05494	−0.04787	−0.04531
0.9	−0.11516	−0.11104	−0.09993	−0.08503	−0.07045	−0.05997	−0.05617

The tabulated quantity is $g(\varepsilon, \gamma) - 1$, where $g(\varepsilon, \gamma)$ is the ratio of the Coulomb energy for the deformed case to the Coulomb energy in the spherical case.

tion, we may replace $|\psi\rangle$ by a simple product of single particle wave functions and obtain

$$E_{\text{coul}} = \frac{1}{2}\sum_{i}\sum_{j}\left\langle \psi(\boldsymbol{r}_i)\,\psi(\boldsymbol{r}_j)\left|\frac{e_2}{r_{ij}}\right|\psi(\boldsymbol{r}_i)\,\psi(\boldsymbol{r}_j)\right\rangle$$

$$= \frac{1}{2}\sum_{i}\sum_{j}\varrho_i(\boldsymbol{r}_i)\left|\frac{e_2}{r_{ij}}\right|\varrho_j(\boldsymbol{r}_j)\,\mathrm{d}^3\boldsymbol{r}_i\,\mathrm{d}^3\boldsymbol{r}_j \tag{14}$$

As a further simplification we assume the nucleus to be a homogenously charged triaxial ellipsoid with semi-axes a, b, and c, respectively. (Lately we have suceeded in also incorporating more general nuclear shapes with $\varepsilon_4 \neq 0$.)

The expression of the Coulomb energy in terms of a, b and c may be written as a simple one-dimensional integral

$$E_{\text{coul}} = \frac{3}{5}(Ze)^2\frac{1}{2}\int_0^\infty \frac{\mathrm{d}\lambda}{\sqrt{(a^2+\lambda)(b^2+\lambda)(c^2+\lambda)}} \tag{15}$$

This expression was kindly simplified for us by S. B. Nilsson [18] and expressed in terms of an ellipsoidal integral:

$$E_{\text{coul}} = \frac{3}{5}(Ze)^2\frac{1}{\sqrt{c^2-a^2}}F(k,\phi_0) \tag{16}$$

where $F(k,\phi)$ is an elliptical integral of first kind and by definition

$$k^2 = \frac{b^2-a^2}{c^2-a^2} \tag{17}$$

and

$$\sin\phi_0 = \frac{\sqrt{c^2-a^2}}{c} \tag{18}$$

Similar expressions are obtained in ref. [19].

If we permit P_4-deformations ($\varepsilon_4 \neq 0$), the general expression of the six-dimensional integral can be reduced no further than to a four-dimensional integral.

Table II. *Same as Table I but for the surface energy*

ε	γ 0°	10°	20°	30°	40°	50°	60°
0.0	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
0.1	0.00180	0.00180	0.00179	0.00177	0.00175	0.00174	0.00174
0.2	0.00741	0.00737	0.00728	0.00715	0.00703	0.00694	0.00690
0.3	0.01714	0.01702	0.01670	0.01626	0.01583	0.01551	0.01539
0.4	0.03146	0.03117	0.03039	0.02932	0.02825	0.02747	0.02719
0.5	0.05102	0.05043	0.04882	0.04664	0.04448	0.04290	0.04232
0.6	0.07672	0.07564	0.07269	0.06871	0.06477	0.06192	0.06088
0.7	0.10989	0.10802	0.10297	0.09619	0.08953	0.08473	0.08299
0.8	0.15255	0.14945	0.14111	0.13002	0.11928	0.11161	0.10885
0.9	0.20824	0.20308	0.18942	0.17165	0.15478	0.14295	0.13871

Table III. *Values of $\varkappa$ and μ employed in the calculations of single-particle levels in the three studied mass regions*

	Protons		Neutrons	
Mass region	$\varkappa_p$	μ_p	$\varkappa_n$	μ_n
$A \simeq 187$	0.0620	0.614	0.0636	0.393
$A \simeq 242$	0.0577	0.650	0.0635	0.325
$A \simeq 298$	0.0534	0.686	0.0634	0.256

At small deformations the Coulomb-energy takes the following form:

$$E_{\text{coul}} = \overset{\circ}{E}_{\text{c}}\left(1 - \frac{4}{45}\varepsilon^2 - \ldots\right) \tag{19}$$

which is thus independent of γ to second order.

$\overset{\circ}{E}_{\text{c}}$ is the Coulomb energy in the spherical case. Further work on γ-degree of freedom together with ε_4-deformations is in progress.

The corresponding expressions for the surface energy are trivial to derive although they lead to fairly complicated analytical formulas which are not presented because of space limitations. We may summarize the expression for the distorition dependent part of the liquid drop contribution to the total energy

$$E_{\text{LD}} = a_{\text{c}}\frac{Z^2}{A^{1/3}}\delta E_{\text{coul}} - a_{\text{s}}(1-\varkappa_{\text{s}}I^2)A^{2/3}\delta E_{\text{surf}} \tag{20}$$

where δE_{coul} and δE_{surf} are the coulomb and surface energy-corrections respectively, tabulated in Tables I and II. Furthermore the isospin dependent term containing $I = (N-Z)/A$ to second order account for the decrease in surface energy with increasing neutron excess [20]. The values employed for the coefficients a_{c} and a_{s} and the surface symmetry coefficient are taken as

$a_{\text{c}} = 0.70531$

$a_{\text{s}} = 17.9439$

$\varkappa_{\text{s}} = 1.7826$

6. Computation of single-particle levels

The adjustable coupling parameters $\varkappa$ and μ are fixed so as to reproduce optimally the observed single-particle level order in both spherical and deformed nuclei [21, 22]. The $\varkappa$ and μ parameters used are given in Table III.

The single-proton and single-neutron energies as a function of γ for the Os and Pt-region investigated are displayed in Figs. 1 and 2. Here $\gamma = 0$ and $\gamma = 60$ corresponds to purely prolate and purely oblate nuclear shapes, respectively. The number of nuclear shells included are $N_{\max} = 14$ (used in the superheavy element region). The fact that we have a two-fold degeneracy, corresponding to Kramer's theorem, makes it certain that the energy-matrix can be split into two independent ones. Thus for example one contains couplings between states with $\Omega = 5/2, 1/2, -3/2$ etc. and the other $\Omega = 3/2, -1/2, -5/2$. They give both the same energy eigenvalues reflecting the above-mentioned time-reversal symmetry.

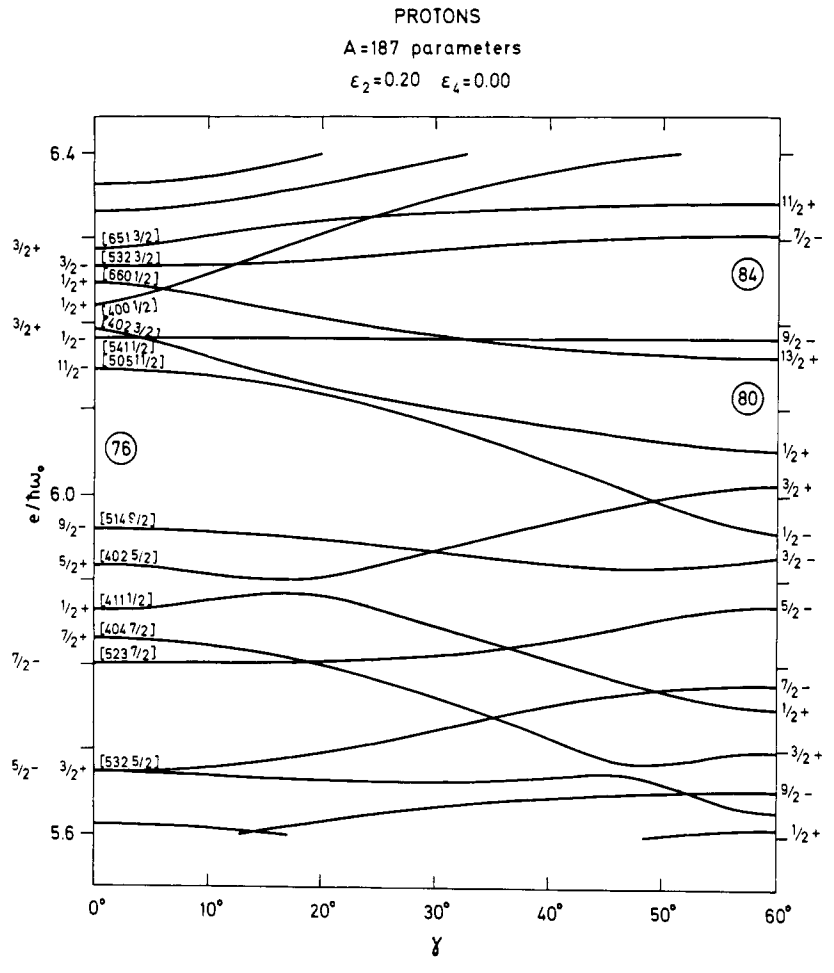


Fig. 1. Single particle levels for protons as a function of γ at $\varepsilon = 0.20$ in the Os–Pt-region. The levels are labeled by their asymptotic quantum numbers $[Nn_z\Lambda\Omega]$. In the prolate and oblate case, where Ω is a good quantum number, the levels are labeled by Ω and parity. Note that on the oblate side Ω refers to the x-axis.

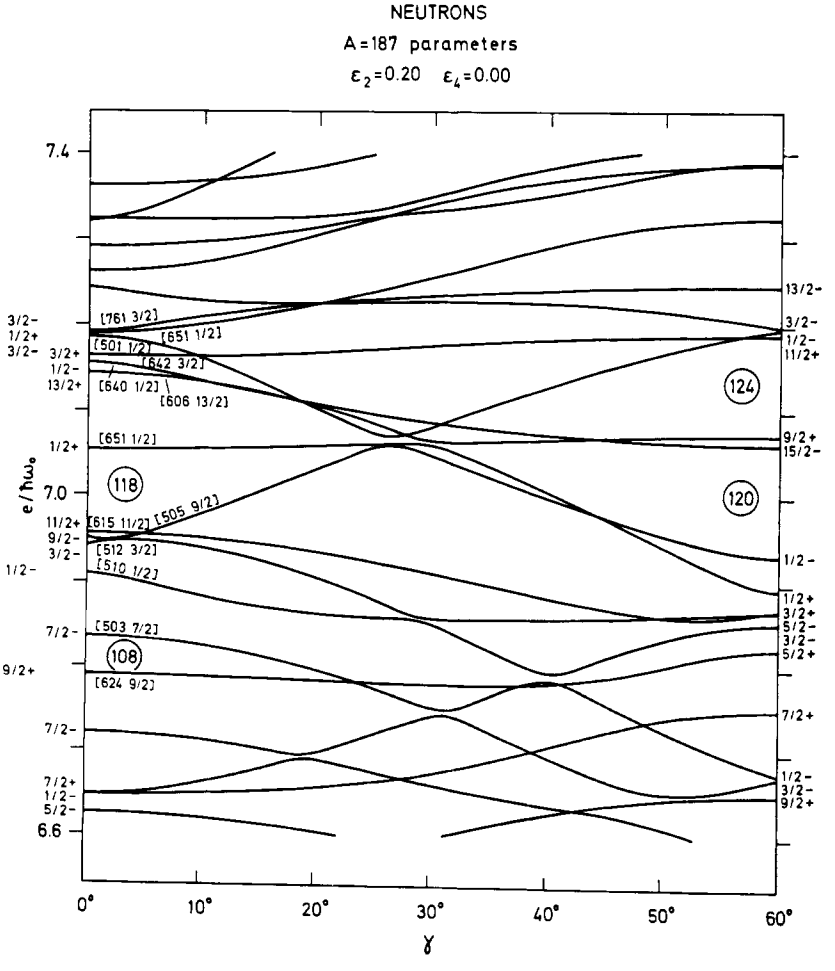


Fig. 2. Same as Fig. 1 for neutrons.

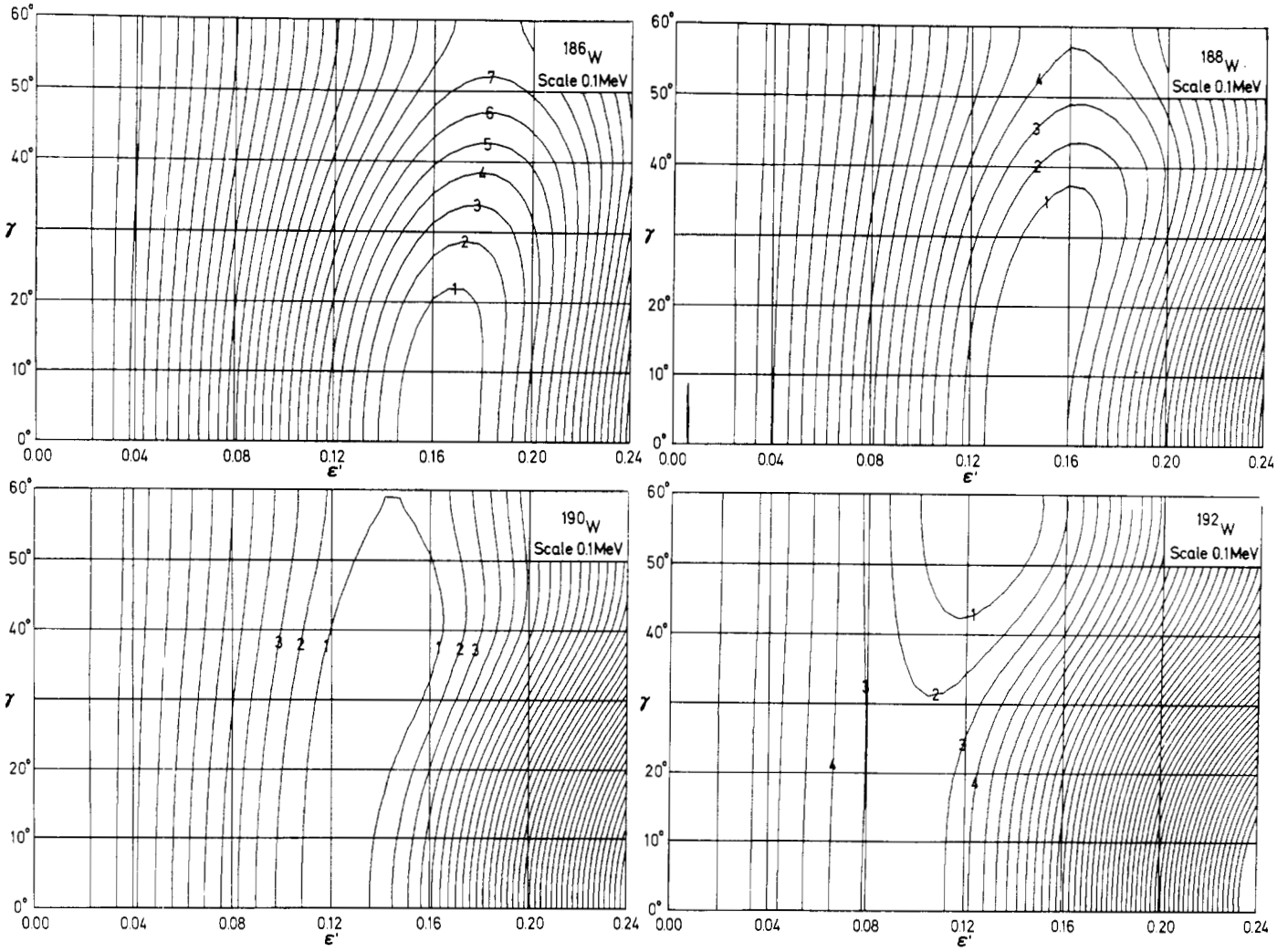


Fig. 3. Potential energy surfaces in (ε, γ)-space for some W-isotopes, calculated with $(G \sim S, \varkappa_s = 1.78)$. The range considered for ε is $0.00 < \varepsilon < 0.24$. Note the prolate-oblate transition. The contour line separation is in units 0.1 MeV.

7. Potential energy surface for even–even nuclei

The nuclear potential energy is thus described by one smooth macroscopic part, the liquid drop part, and one microscopic part, which latter accounts for the fluctuations associated with the shell structure in the nucleus. Thus we have the total potential energy

$$\mathcal{E}_{\text{tot}} = E_{\text{LD}} + (\delta E_{\text{shell}})_{\text{p+n}} + (\delta E_{\text{pair}})_{\text{p+n}}$$

$$\delta E_{\text{shell}} + \delta E_{\text{pair}} = \sum_\nu 2\varepsilon_\nu v_\nu^2 - \frac{\Delta^2}{G} - G\left(\sum_\nu v_\nu^4 - \sum' 1\right) - E(g) - \langle E_{\text{pair}} \rangle \tag{21}$$

where $E(g) = \int^{E_F} 2eg(e)\,de$ is the smooth total single-particle energy. The smoothed level density $g(e)$ can be written

$$g(e) = \frac{1}{\gamma\sqrt{\pi}} \sum_\nu f_{\text{corr}}\, e^{-\left(\frac{e-e_\nu}{\gamma}\right)^2} \tag{22}$$

the microscopic part is treated in accordance with the generalized Strutinsky prescription [23–26], where the shell correction energy

$$E_{\text{shell}} = \sum_\nu \varepsilon_\nu - \widetilde{\sum_\nu \varepsilon_\nu} \tag{24}$$

In the present work $\widetilde{\Sigma_\nu \varepsilon_\nu}$ is calculated with a sixth order smearing polynominal with the smearing parameter $\gamma = 1.2\hbar\omega_0$. The pairing matrix elements are taken as

$$(G \mp A)_{\substack{\text{p} \\ \text{n}}} = g_0 \mp g_1 \frac{N-Z}{A} \tag{24}$$

where we have used $g_0 = 19.2$ MeV and $g_1 = 7.4$ MeV. We also assume the basic matrix elements to be proportional to the surface area. The quoted numbers thus refer to spherical shape. For deformed shape we have assumed G being proportional to the surface area. To reproduce over a large range in A the empirical odd-even mass difference Δ, the cut-off energy is chosen so that we only include $\sqrt{15Z}$ or $\sqrt{15N}$ states above and below the proton or neutron Fermi-level.

7.1. *$A \simeq 190$ region*

In Fig. 3, 4 and 5 some representative potential energy surfaces for isotopes of W, Os and Pt are presented. As mentioned these diagrams were calculated under the assumption of G being proportional to the surface area S, furthermore the surface symmetry coefficient $\varkappa_s$ was equal to 1.78 $(G \sim S,\ \varkappa_s = 1.78)$. The general trend apparent is that of a continuous shift of the minimum form prolate to oblate shape when A increases. Between nearly purely prolate and oblate shapes there occur equilibrium values of $\gamma \simeq 20°$–$40°$ for intermediate A-values. For W prolate equilibrium

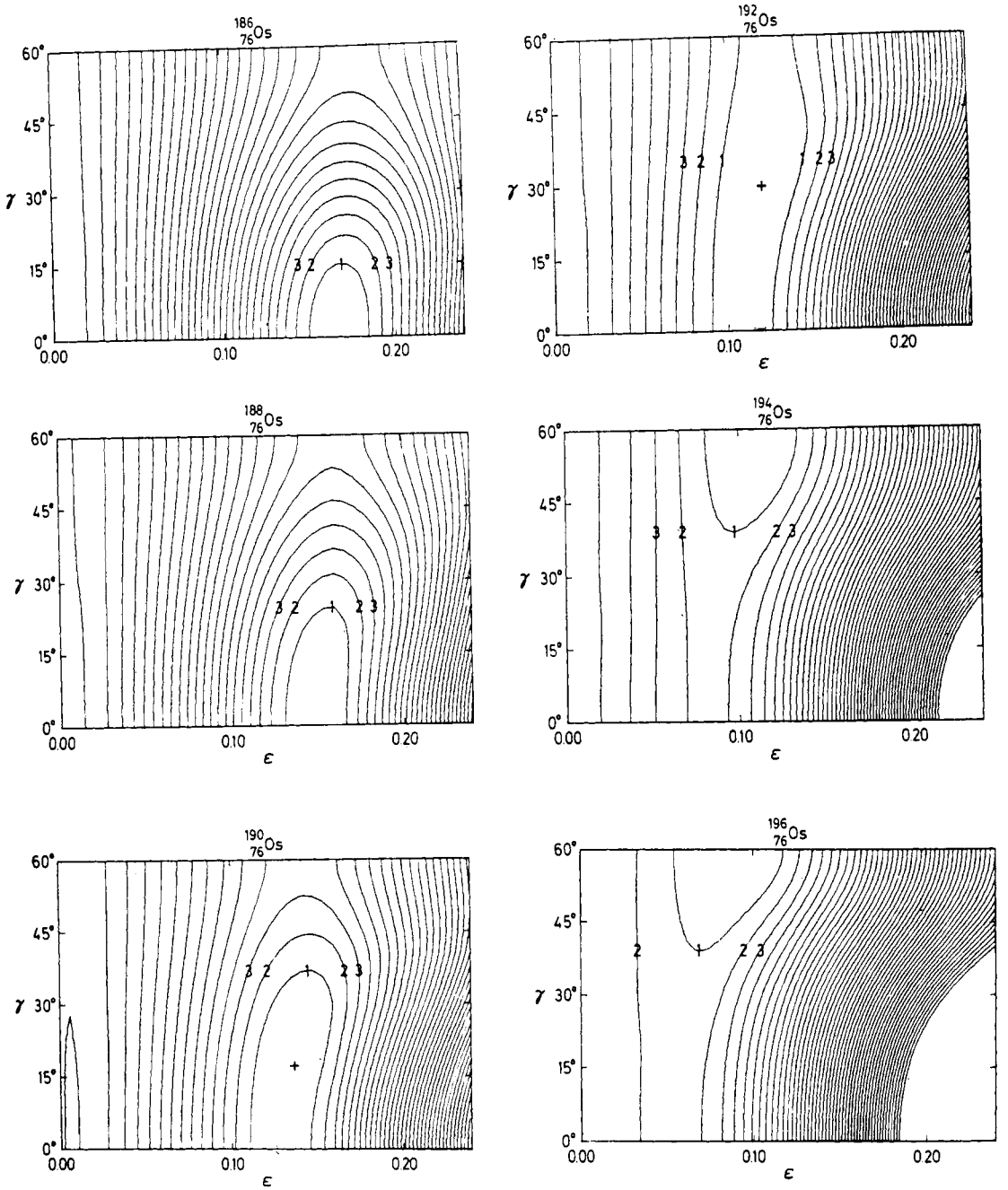


Fig. 4. Same as Fig. 3 for Os-isotopes.

deformations occur for $A < 186$, while for $A = 188$, 190 the minimum in the potential energy surface appears in the γ-plane. Again, for $A > 192$ pure oblate shapes are obtained. Very similar results are obtained for Os and Pt where the intermediate γ-values of $\gamma \simeq 20°$ and $\gamma \simeq 40°$ occur for $A = 190$ and 192, respectively. This is in good agreement with results obtained in ref. [27].

In Figs. 6 and 7 a sequence of W, Os and Pt isotopes are depicted with a range of deformation extended to $\varepsilon = 0.6$ and a contour line separation of 0.5 MeV. A secondary energy minimum for $\varepsilon \simeq 0.4$ is apparent.

In Figs. 8 and 9 potential energy surfaces for some elements with mass $A \simeq 200$ are presented, showing the secondary minimum in this mass region already predicted by Gustafsson et al. [28]. These secondary minima are stable to γ-deformations, although the barrier towards spherical shape is lowered by up to 2 MeV by the inclusion of γ. The electromagnetic decay back to the spherical ground state should be associated with an extra hindrance due to this change in shape. It should thus be possible to establish experimentally the existence of this shape isomer predicted. The nuclides ^{200}Hg or ^{200}Pb appear to be particularly favourable cases (Fig. 9).

To illustrate the effect of the γ-coordinate on the energy surfaces for various nucleon numbers, the shell correction energy and the shell correction plus pairing energy for protons and neutrons respectively have been plotted in Figs. 10 and 11. In Fig. 11 there is apparent a strong oblate tendency to produce oblate shapes for $N > 112$ ($N = 112$ corresponds to a saddle point).

7.2. *Actinide region*

Figs. 12 and 13 present the nuclear potential energy surfaces for some representative actinide elements. The calculations have been performed in the (ε, γ)-space with $\varepsilon_4 = 0$ as in the mass region below $A = 208$. The nuclei exhibited are deformed in their ground state and have $\varepsilon \simeq 0.2$. The inclusion of the γ-coordinate has the effect of reducing the first barrier for nuclei with $Z > 94$ while the Th, U, Pu isotopes are less effected. The γ-distortion of the first saddle is shown in Fig. 14. The general trend is that the γ-distortion is minimal for neutron numbers around $N = 152$–154.

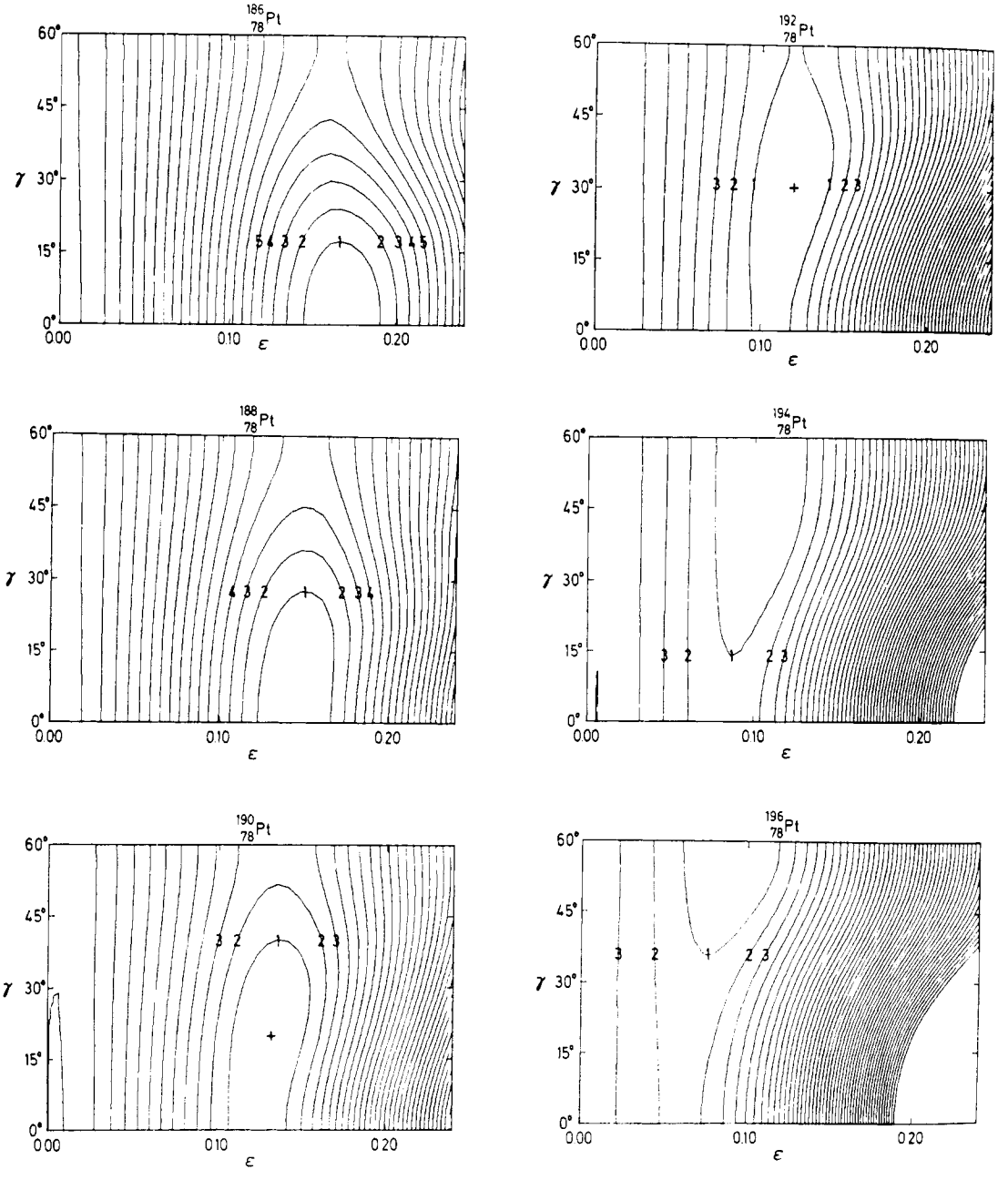


Fig. 5. Same as Fig. 3 for Pt-isotopes. The crosses corresponds to the lowest value of the potential energy surface.

In Fig. 15 the reduction due to γ-asymmetry is plotted as a function of neutron-number. Here the minimum reduction occurs from neutron numbers $N = 154$–156. The connection between the theoretical γ-asymmetry and the observed fission fragment angular distribution is discussed in ref. [29]. The two alternate choices of parameters have been made in the calculations namely a surface-dependent pairing $G \sim S$ and $\varkappa_s = 1.78$ in one variant and a constant pairing matrix element $G = \text{const.}$ and $\varkappa_s = 2.53$ according to the other variant. In some cases the reduction of the energy of the first saddle is about 1.5 MeV. As shown in Fig. 16 the calculated energy of the first saddle, relative to the ground state as a function of mass number A, is thus in better agreement with experimental barrier heights [30, 31]. Finally for the actinides, the shell correction plus pairing energies are shown in Fig. 17*a* (protons) and *b* (neutrons) at $\varepsilon = 0.42$ (corresponding to the first peak) under the restriction $\gamma = 0°$. For the cases that the introduction of the γ-degree of freedom leads to a lowering of the saddle point, the saddle is usually moved to $\gamma \simeq 30°$.

In Fig. 18 the nucleus ^{230}Th is displayed in a plot where particular emphasis is put on finer details of the energy surface near the minimum. This plot shows a triaxial isomeric minimum in addition to the prolate state. The separating barrier between the minima is, however, very low. Experimental spectra for light Th and U nuclei as well as transition probabilities there give strong evidence of two rotational band systems, one built in such a triaxial minimum [32] in addition to one built on a prolate minimum.

7.3. *The superheavy-element region*

In later years attention has been directed to the expected island of superheavy elements. Of special interest are nuclei around ^{298}114. Thus the latter corresponds to double closed shells similar to those near ^{208}Pb. Thus the level spectrum exhibits gaps for $Z = 114$ and $N = 184$. These gaps have been obtained in several calculations based on various nuclear potentials. The halflives in this region of elements are largely determined by the fission process and heights of the fission barriers are decisive. In determining those the γ-degree of freedom appears highly important.

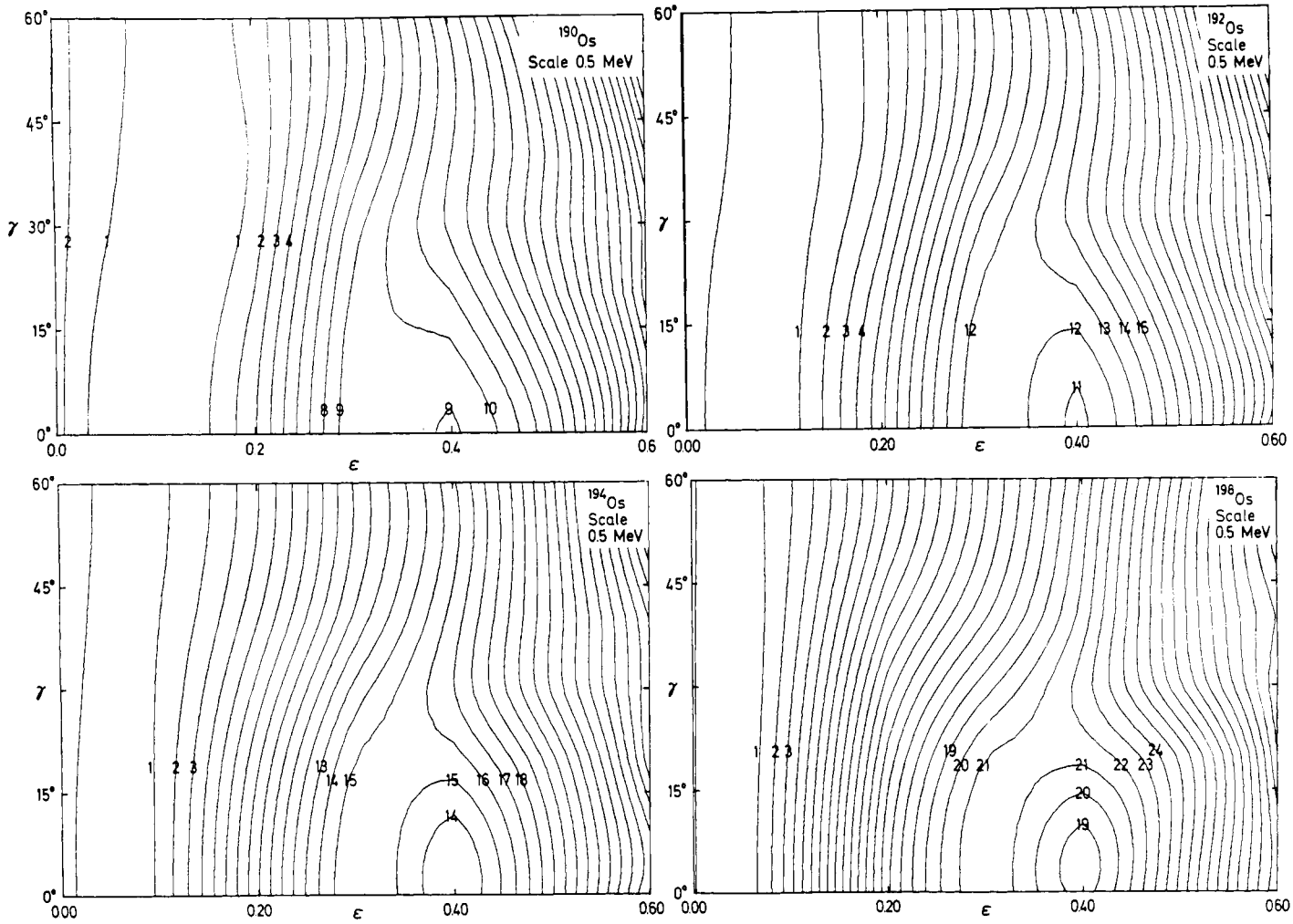


Fig. 6. Potential energy surfaces for some Os-isotopes, where the range for ε is extended to $\varepsilon = 0.60$ and the contour line separation is 0.5 MeV.

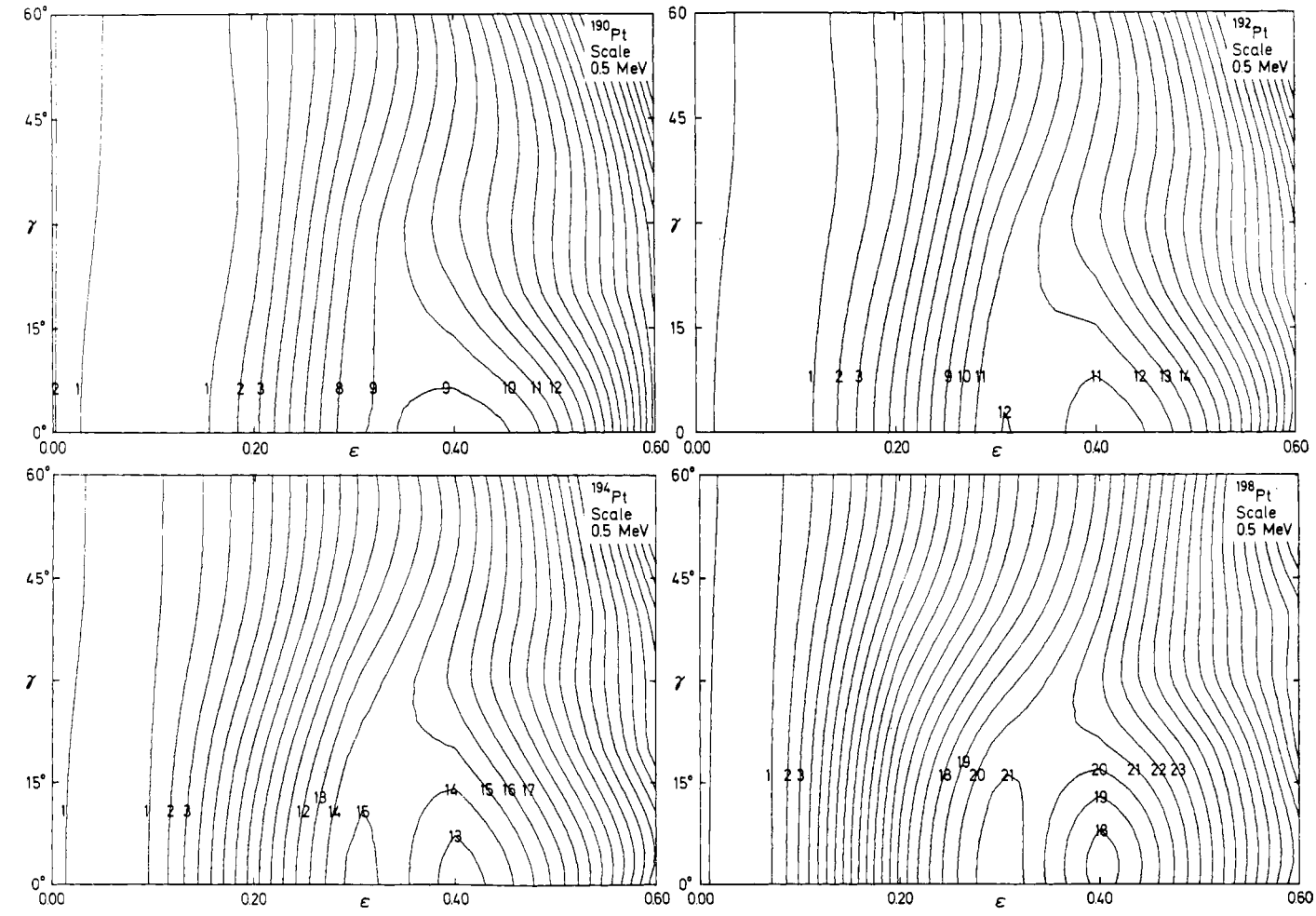


Fig. 7. Same as Fig. 6 for Pt isotopes.

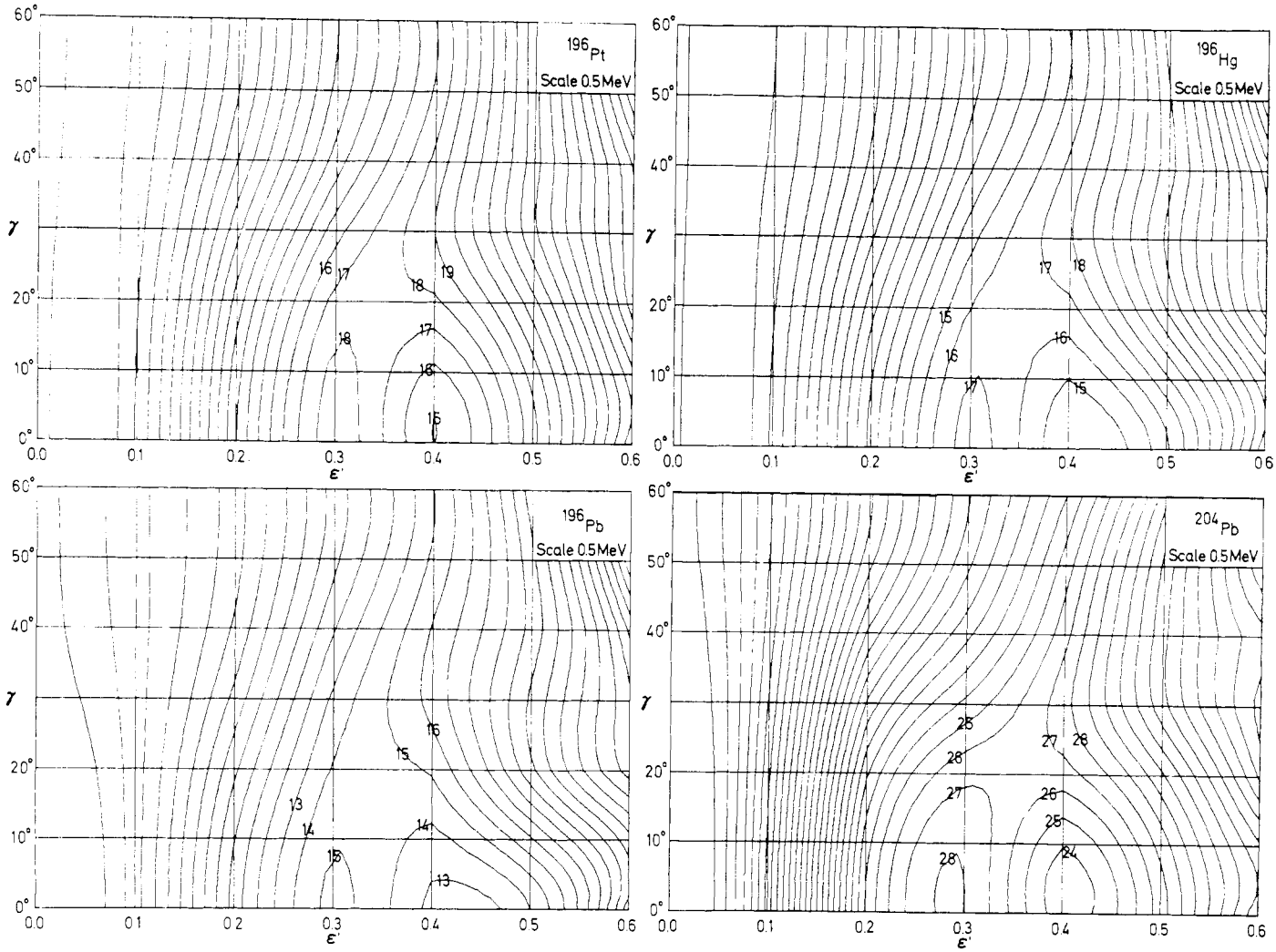


Fig. 8. Potential energy surfaces in terms of ε and γ for some $A \cong 200$ elements. Note the secondary minima at $\varepsilon \cong 0.45$ and the reduction of the barrier due to the γ-degree of freedom. The contour line separation is 0.5 MeV.

From these calculations the potential energy surface for some superheavy elements are obtained in Figs. 19 and 20. For the nuclei in question the ground state generally occurs at zero deformation, except for some neutron deficient cases (for example $^{284}114$) and the second minima fall at about $\varepsilon = 0.4$–0.5. The effect of inclusion of axial asymmetry is to lower the first barrier by up to 1.5 MeV. The results are collected in Table V, where the γ-distortion at the first saddle and the energy reduction, ΔE_γ due to γ are tabulated.

From Table V we obtain the very interesting result that the saddle point is located in the γ-plane at about $\gamma = 30°$ for all nuclei considered. The barrier heights for some nuclei are shown in Fig. 21. The theoretical barrier heights including ε_4 but without the γ-degree of freedom are taken from ref. [33]. The nucleus

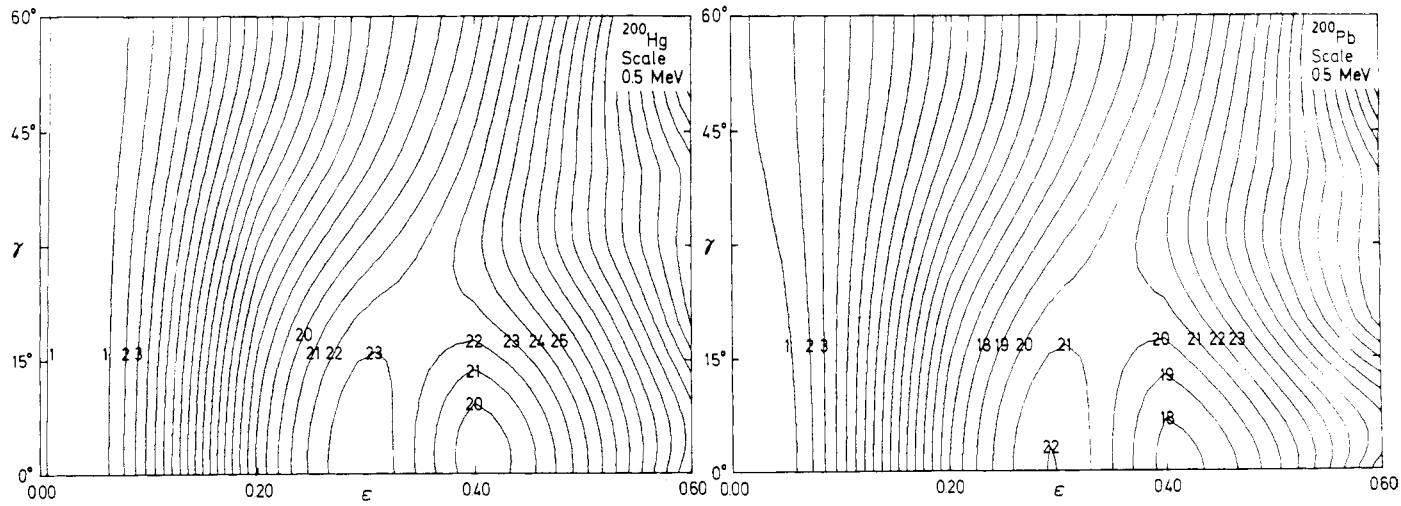


Fig. 9. Same as Fig. 8.

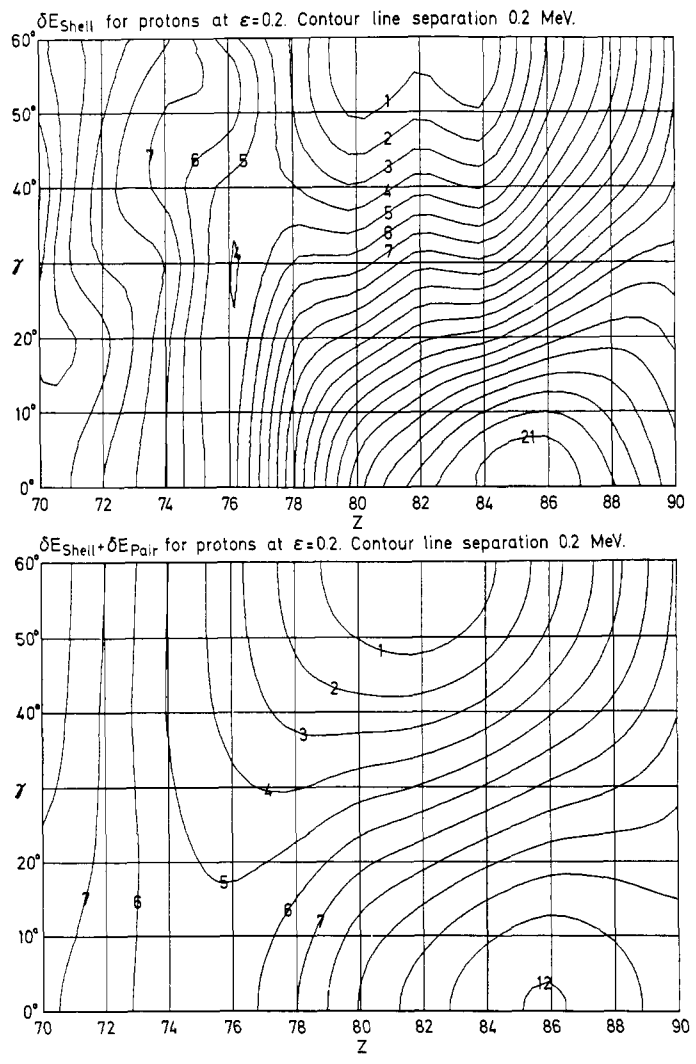


Fig. 10. Shell correction and shell plus pairing energy, respectively. For protons at $\varepsilon = 0.2$ calculated with ($G \sim S, \varkappa_s = 1.78$) and $A \cong 187$ parameters. Note the oblate preference for larger Z-values. The pairing energy has the effect of smoothing the energy surface.

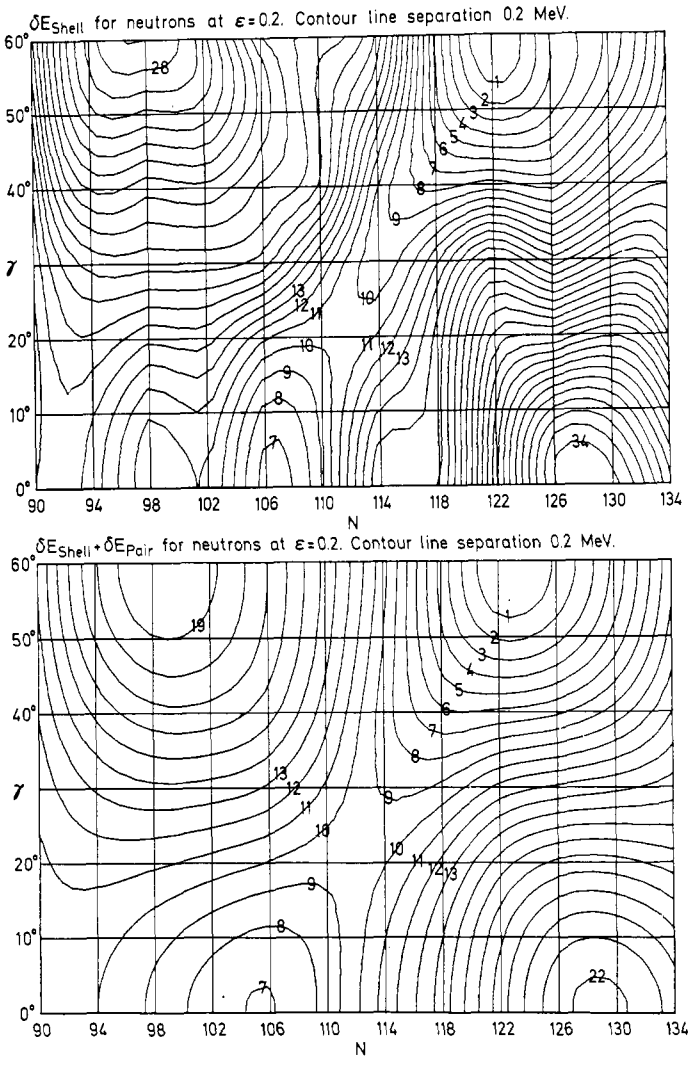


Fig. 11. Same as Fig. 10 for neutrons and $\varepsilon = 0.2$. Note the saddle point in the energy surface for neutron number $N = 112$. For $N > 112$ oblate shapes are favoured.

Table IV. *Values of γ and ΔE_{saddle} at first saddle point for actinides (using $G \sim S$, $\varkappa_s = 1.78$)*

Nucleus	γ_{saddle}	ΔE_{saddle} (MeV)	Nucleus	γ_{saddle}	ΔE_{saddle} (MeV)
^{232}Pu	0°	0	^{250}Cf	4°	0.9
^{234}Pu	0°	0	^{252}Cf	2°	0.4
^{236}Pu	5°	0	^{254}Cf	8°	0.2
^{238}Pu	10°	0.1	^{256}Cf	9°	0.5
^{240}Pu	10°	0.2	^{244}Fm	18°	1.2
^{242}Pu	9°	0.4	^{246}Fm	16°	1.3
^{244}Pu	7°	0.3	^{248}Fm	14°	1.4
^{246}Pu	3°	0.2	^{250}Fm	10°	1.4
^{238}Cm	17°	0.3	^{252}Fm	5°	1.3
^{240}Cm	14°	0.4	^{254}Fm	7°	0.8
^{242}Cm	12°	0.6	^{256}Fm	10°	0.4
^{244}Cm	10°	0.7	^{258}Fm	10°	0.7
^{246}Cm	8°	0.8	^{260}Fm	10°	1.0
^{248}Cm	3°	0.5	^{250}No	21°	1.5
^{250}Cm	0°	0.1	^{252}No	20°	1.6
^{252}Cm	6°	0.1	^{254}No	8°	1.6
^{254}Cm	9°	0.3	^{256}No	9°	1.1
^{242}Cf	16°	0.8	^{258}No	11°	0.9
^{244}Cf	14°	1.0	^{260}No	11°	1.0
^{246}Cf	12°	1.1	^{262}No	10°	1.3
^{248}Cf	9°	1.1	^{264}No	9°	1.5

The quantity ΔE_{saddle} is the energy reduction due to the inclusion of γ.

^{298}114 still retains the highest barrier even when γ is included. At this stage, no predictions can be made as to the reduction of the fission half-lives without a dynamical calculation. Furthermore the components of the mass tensor relevant to the gamma degree of freedom are not available. As a rough estimate we may expect a reduction in half-lives of up to a few powers of ten.

8. Conclusions

The effect of the inclusion of the degree of freedom of axial asymmetry is found to reproduce the prolate-oblate transition behaviour in the Os–Pt-region satisfactorily when ε_4 is neglected. In the $A \simeq 200$-region the barrier separating the secondary minimum from the ground state is found to be reduced up to 2 MeV for some nuclei. Thus, from the present standpoint only in fortunate case should this type of isomeric states be observable in this mass region.

We furthermore find that the inner one of the fission barriers in the actinide region is reduced by up to 1.5 MeV, thus improving the agreement with empirically estimated barrier heights. The second barrier is stable to axially asymmetric distortions, but on

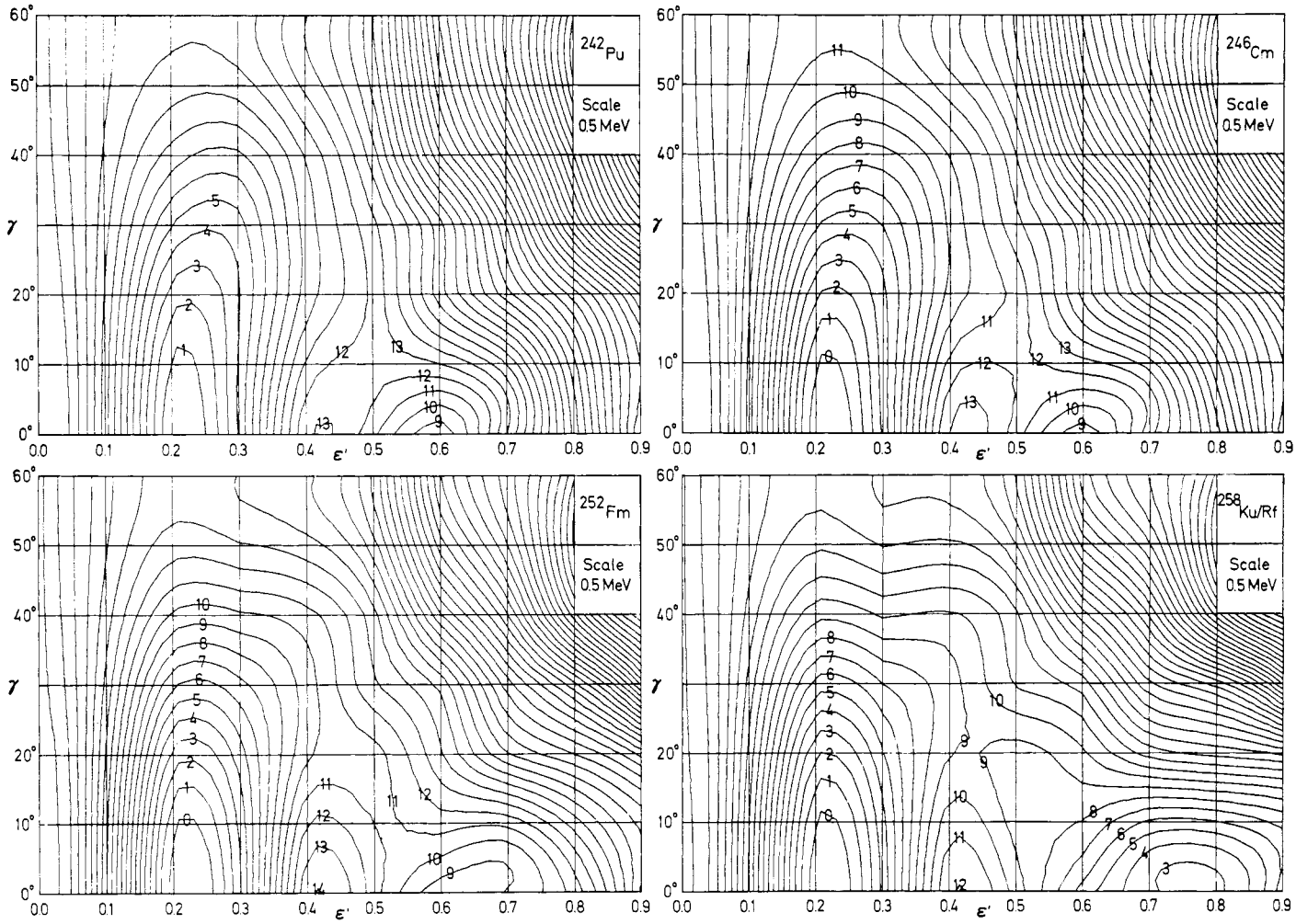


Fig. 12. Potential energy surfaces for some elements in the actinide region calculated with $A \simeq 242$-parameters and ($G \sim x$, $\varkappa_s = 1.78$). Note the energy reduction due to γ at the first saddle point. The contour line separation is in units of 0.5 MeV.

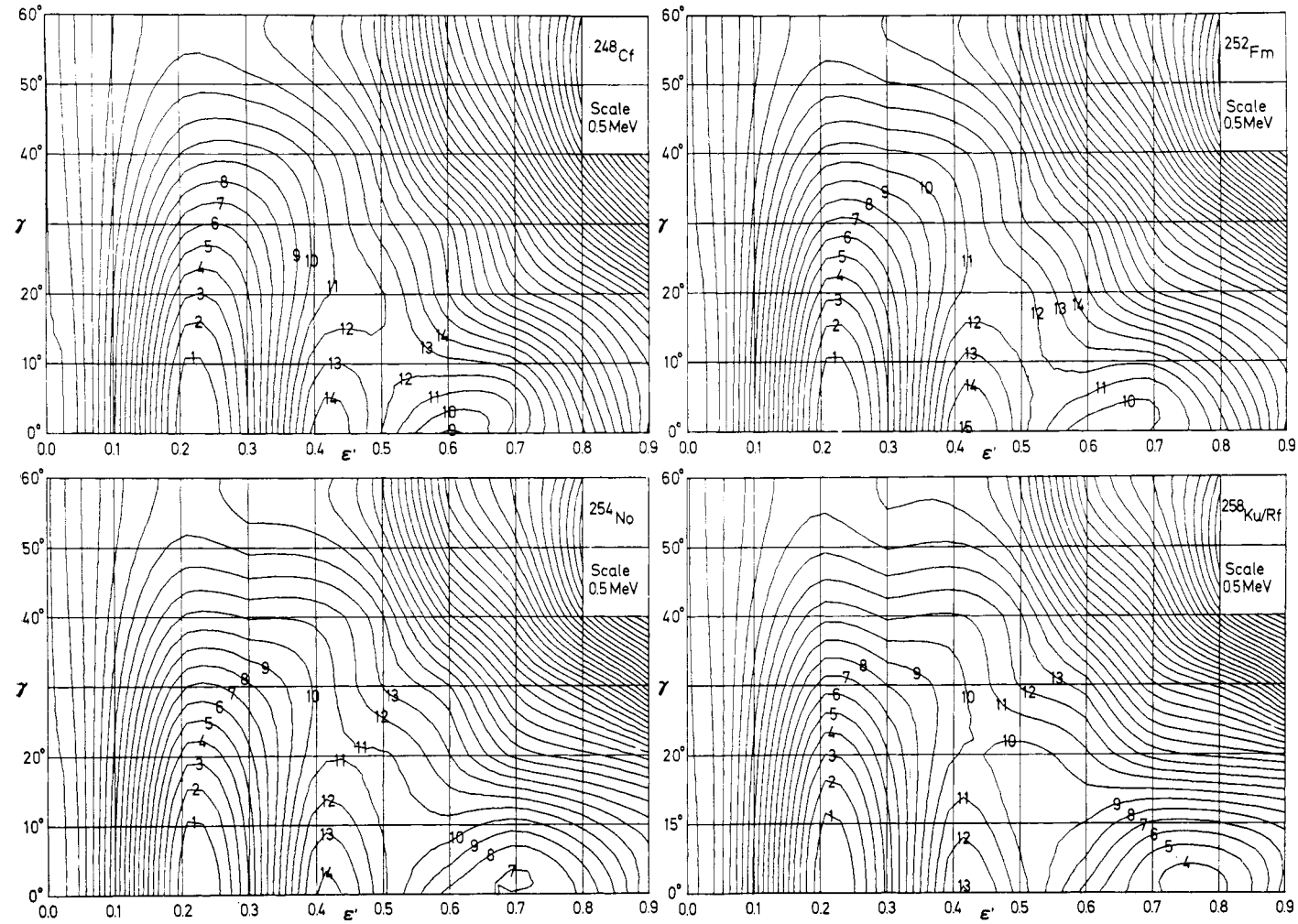


Fig. 13. Same as Fig. 12 but for some other actinide elements.

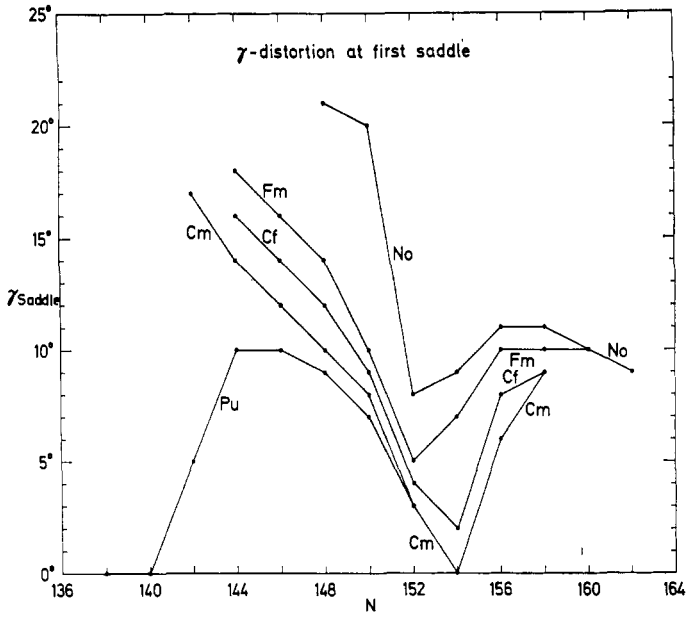


Fig. 14. γ-distortions of the first saddle as a function of neutron number, calculated with ($G\sim S, \varkappa_s = 1.78$).

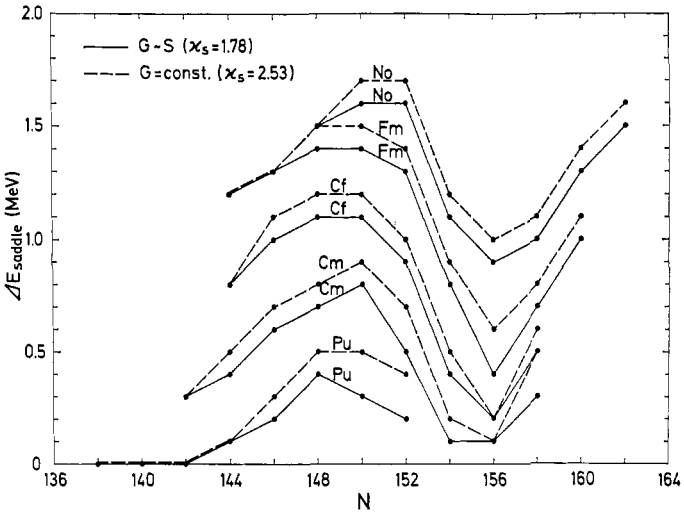


Fig. 15. The energy reduction due to γ-degree of freedom of the first saddle as a function of neutron number calculated in the two alternative variants ($G\sim S, \varkappa_s = 1.78$) and ($G = \text{const.}, \varkappa_s = 2.53$) respectively. The least reduction occurs at $N = 152$–154.

the other hand, unstable to reflection asymmetric degrees of freedom.

Finally, in the superheavy region of elements, the fission barrier reduction for $Z < 114$ and $N < 114$ is generally modest and less than 1 MeV. On the other hand above these neutron ad proton numbers the effect is somewhat larger.

As the complete mass parameters associated with the γ-degree of freedom are not presently available, no detailed estimates of the corresponding correction factors for the fission half-lives can be made.

Acknowledgements

The author is very much indebted to Professor S. G. Nilsson for suggesting this problem and for many helpful and informative discussions. Thanks are also due to my colleagues in the Lund group and to Professors R. Sheline and Z. Szymanski, Nordita guest professors, for valuable comments. The author also wishes to thank Mr L. Nilsson who has done an excellent job in drafting the figures.

Table V. *Same as Table IV but for superheavy elements*

Nucleus	γ_{saddle}	ΔE_γ (MeV)
$^{288}110$	21°	0.2
$^{290}110$	23°	0.3
$^{292}110$	25°	0.5
$^{294}110$	26°	0.5
$^{296}110$	27°	0.7
$^{290}112$	23°	0.3
$^{292}112$	26°	0.5
$^{294}112$	27°	0.6
$^{296}112$	29°	0.8
$^{298}112$	29°	0.9
$^{292}114$	27°	0.6
$^{294}114$	27°	0.9
$^{296}114$	27°	1.1
$^{298}114$	28°	1.3
$^{300}114$	30°	1.4
$^{294}116$	30°	1.1
$^{296}116$	30°	1.4
$^{298}116$	30°	1.4
$^{300}116$	30°	1.6
$^{302}116$	30°	1.5

Appendix I

The transformation to tri-axially stretched coordinates can be worked out as follows. It is easy to derive the following commutator relations.

$$[\Delta_\xi, \xi^2 + \eta^2 - 2\zeta^2] = 4\xi\frac{\partial}{\partial\xi} + 4\eta\frac{\partial}{\partial\eta} - 8\zeta\frac{\partial}{\partial\zeta} \tag{25}$$

and

$$[\Delta_\xi, [\Delta_\xi, \xi^2 + \eta^2 - 2\zeta^2]] = 8\left(\frac{\partial^2}{\partial\xi^2} + \frac{\partial^2}{\partial\eta^2} - 2\frac{\partial^2}{\partial\zeta^2}\right) \tag{26}$$

Furthermore

$$[\varrho^2, [\Delta_\xi, \xi^2 + \eta^2 - 2\zeta^2]] = -8(\xi^2 + \eta^2 - 2\zeta^2) \tag{27}$$

Similarly one can show

$$[\Delta_\xi, \xi^2 - \eta^2] = 4\xi\frac{\partial}{\partial\xi} - 4\eta\frac{\partial}{\partial\eta} \tag{28}$$

and

$$[\Delta_\xi, [\Delta_\xi, \xi^2 - \eta^2]] = 8\left(\frac{\partial^2}{\partial\xi^2} - \frac{\partial^2}{\partial\eta^2}\right) \tag{29}$$

$$[\varrho^2, [\Delta_\xi, \xi^2 - \eta^2]] = -8(\xi^2 - \eta^2) \tag{30}$$

From the evaluation of these commutators between states of $N' \neq N$ and $N' = N$ respectively the formulas quoted in the text are easily proven.

Appendix II

The pairing force

With the inclusion of the pairing force, corresponding to the short range components of the nucleon–nucleon force, the total

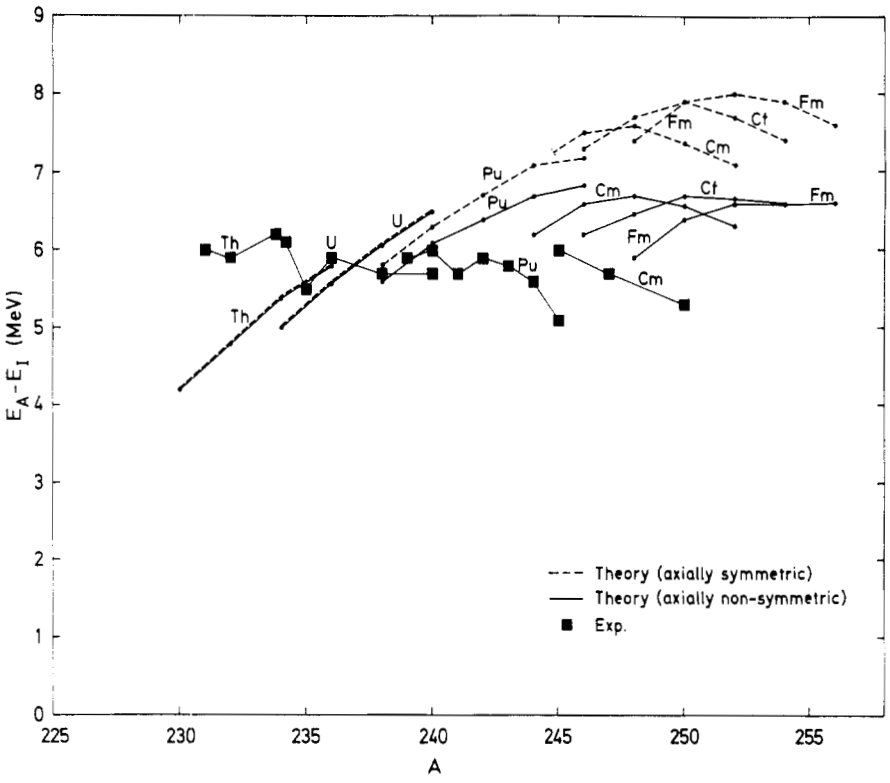


Fig. 16. The calculated inner barrier height with and without the γ-degree of freedom compared with experimental barrier heights.

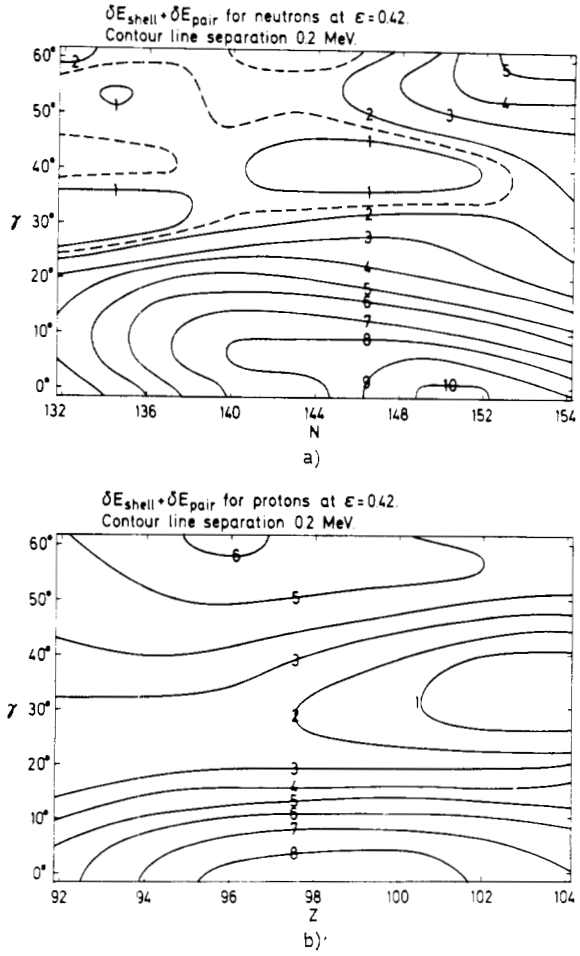


Fig. 17a. Shell correction including pairing energy for neutrons in the actinide region at $\varepsilon = 0.42$ (corresponding to the inner barrier) with γ taken into account. The energy gain due to γ is about 2 MeV.

Fig. 17b. Same as Fig. 21*a* for protons. The dashed lines correspond to a contour line separation of 0.1 MeV. The energy gain due to γ is about 1 MeV for protons.

Hamiltonian may be written, in the language of second-quantization

$$H = \sum_{\nu} \varepsilon_{\nu} (a_{\nu}^{+} a_{\nu} + a_{\bar{\nu}}^{-} a_{\bar{\nu}}) - G \sum_{\mu, \nu} a_{\mu}^{+} a_{\bar{\mu}}^{+} a_{\bar{\nu}} a_{\nu} \tag{31}$$

where ε_{ν} are the single particle energies, and G is the pairing matrix element. In the BCS method (17) the ground state is approximated by

$$|\eta_0\rangle = \prod_{\nu} (u_{\nu} - v_{\nu} a_{\nu}^{-} a_{\bar{\nu}}^{-}) |0\rangle \tag{32}$$

where the pairing factors u_{ν} and v_{ν} are free parameters to be determined fulfilling a normalisation condition

$$u_{\nu}^2 - v_{\nu}^2 = 1 \tag{33}$$

and where $|0\rangle$ stands for particle vaccuum. Under these assumptions the average number of particles are given by

$$N = 2 \sum_{\nu} v_{\nu}^2 \tag{34}$$

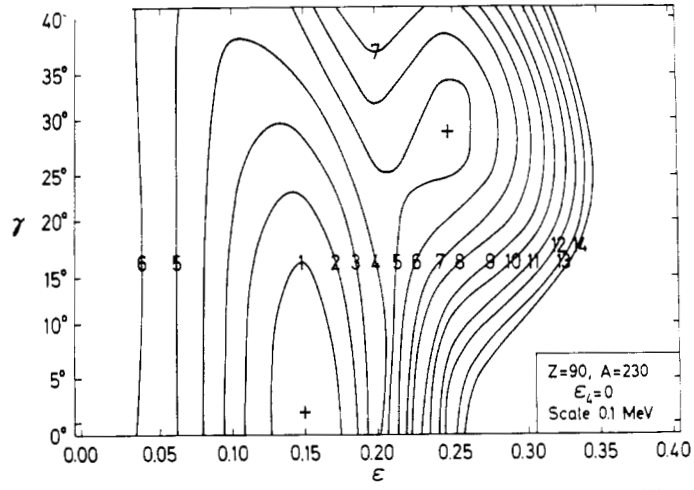


Fig. 18. Potential energy surface in terms of ε and γ (and $\varepsilon_4 = 0.0$) for $^{230}_{90}$Th. Note the soft triaxial minimum at $\gamma \cong 30°$.

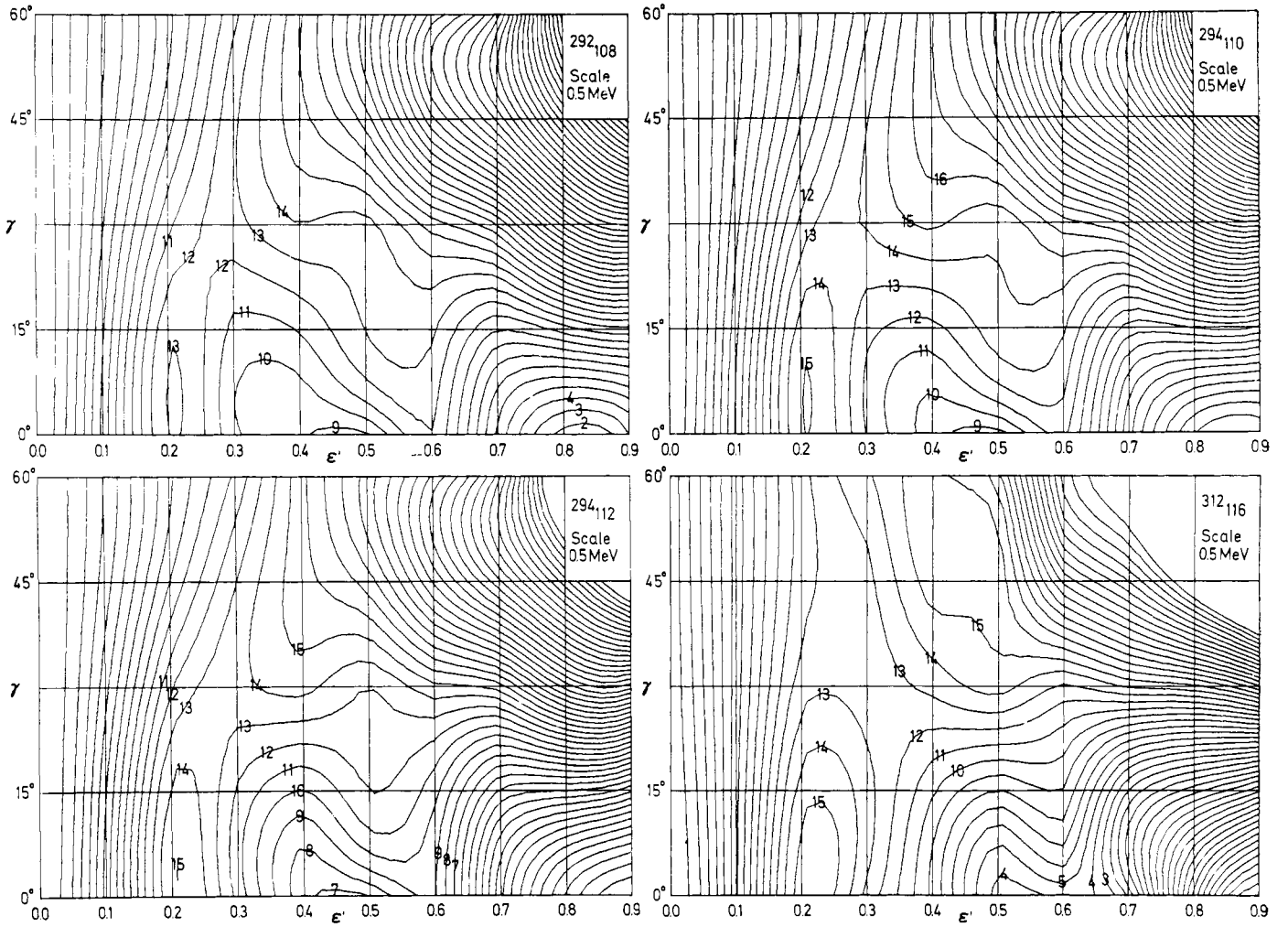


Fig. 19. Potential energy surfaces for some elements in the superheavy region. The γ-value at the saddle point is about $\gamma = 25°$–$30°$ in general.

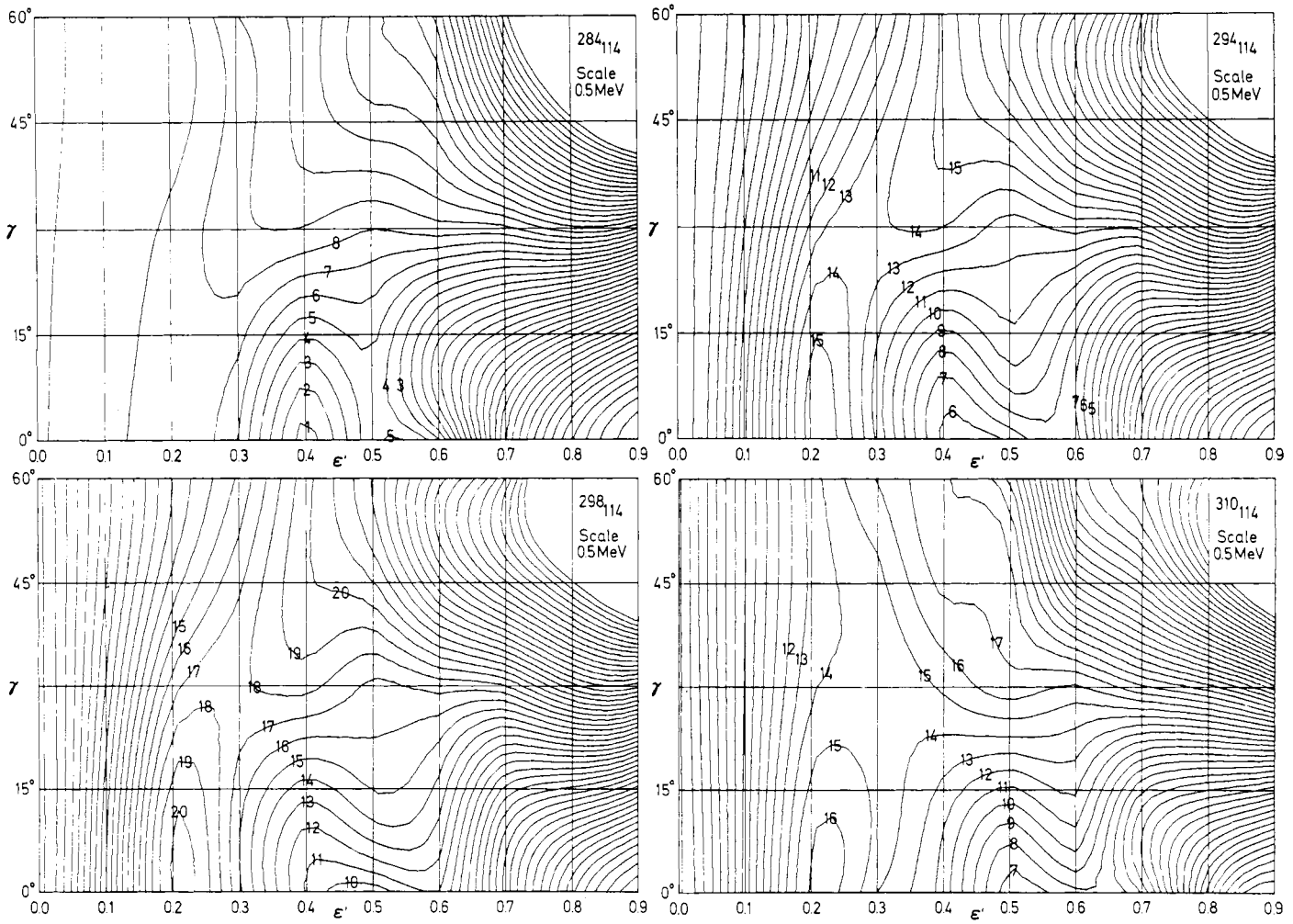


Fig. 20. Same as Fig. 19 for some other superheavy elements.

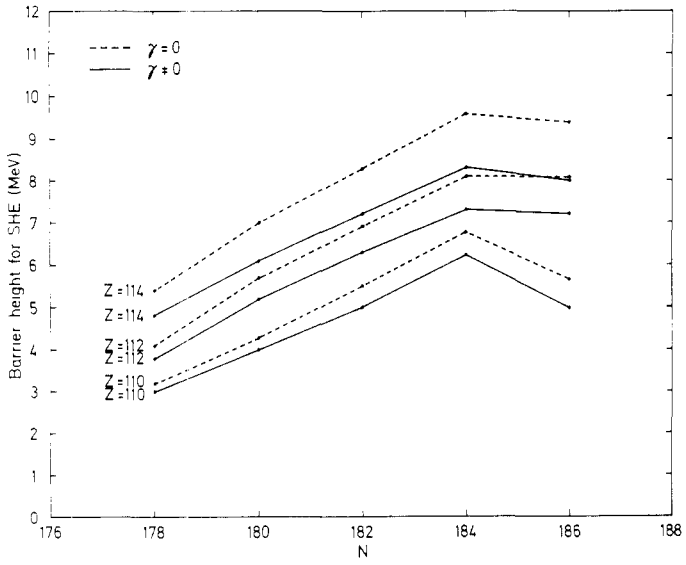


Fig. 21. The barrier heights of some superheavies with inclusion of the γ-degree of freedom. The highest barrier is still encountered for $^{298}_{184}114$.

As the particle number of particles are conserved only on the average, the energy in the BCS-approximation is determined by variational calculus of the Hamiltonian

$$\tilde{H} = H - \lambda N \tag{35}$$

where λ represent the Fermi-energy level. The energy is given by

$$E = \sum_{\nu} 2\varepsilon_{\nu} v_{\nu}^2 - \frac{\Delta^2}{G} - G\sum_{\nu} v_{\nu}^4 \tag{36}$$

The gap parameter Δ is given by

$$\Delta = G\sum_{\nu} u_{\nu} v_{\nu} \tag{37}$$

and the pairing factors will be

$$\begin{aligned} u_{\nu}^2 &= \frac{1}{2}\left(1 - \frac{\varepsilon_{\nu} - \lambda}{\sqrt{(\varepsilon_{\nu} - \lambda)^2 + \Delta^2}}\right) \\ v_{\nu}^2 &= \frac{1}{2}\left(1 - \frac{\varepsilon_{\nu} - \lambda}{\sqrt{(\varepsilon_{\nu} - \lambda)^2 + \Delta^2}}\right) \end{aligned} \tag{38}$$

From these equations we can establish the gap equation

$$\frac{2}{G} = \sum_{\nu} \frac{1}{\sqrt{(\varepsilon_{\nu} - \lambda)^2 + \Delta^2}} \tag{39}$$

From this equation under the constraint that the particle number is conserved on the average we obtain the Fermi energy level λ.

Since the single particle levels ε_{ν} are known, the pairing energy is then given by eq. (36).

References

1. Möller, P., Nucl. Phys. **A192**, 529 (1972).
2. Pashkevich, V. V., Nucl. Phys. **A133**, 400 (1969).
3. Schultheiss, H. and Schultheiss, R., Phys. Lett. **34B**, 245 (1971).
4. Schultheiss, H. and Schultheiss, R., Phys. Lett. **35B**, 296 (1971).
5. Götz, U., Pauli, H. C. and Alder, K., Phys. Lett. **38B**, 274 (1972) and private communication.
6. Larsson, S. E., Ragnarsson, I. and Nilsson, S. G., Phys. Lett. **38B**, 269 (1972).
7. Bohr, A., Mat. Fys. Medd. Dan. Vid. Selsk. **26**, No. 14 (1952).
8. Bohr, A. and Mottelson, B., Mat. Fys. Medd. Dan. Vid. Selsk. **27**, No. 16 (1953).
9. Kumar, K. and Baranger, M., Nucl. Phys. **A62**, 113 (1965).
10. Kumar, K. and Baranger, M., Nucl. Phys. **A110**, 490 (1968).
11. Kumar, K. and Baranger, M., Nucl. Phys. **A110**, 529 (1968).
12. Johnson, A. and Foucher, R., to be published.
13. Johnson, A., thesis, Faculté de Science de Paris, Paris, 1971.
14. Newton, T. D., Can. J. Phys. **38**, 70 (1960).
15. Nilsson, S. G., Dan. Mat. Fys. Medd. Selsk. **29**, (1955).
16. Nilsson, S. G., Tsang, C. F., Sobiczewski, A., Szymański, Z., Wycech, S., Gustafson, C., Lamm, I. L., Möller, P. and Nilsson, B., Nucl. Phys. **A131**, 1 (1969).
17. Fraser, R., Grumann, J. and Greiner, W., Phys. Lett. **35B**, 483 (1971).
18. Nilsson, S. B., private communication.
19. Götz, U., Pauli, H. C. and Alder, K., Nucl. Phys. **A175**, 481–494 (1971).
20. Myers, W. D. and Swiatecki, W. J., Nucl. Phys. **81**, 1 (1966).
21. Nilsson, S. G., Nix, J. R., Sobiczewski, A., Szymański, Z., Wycech, S., Gustafson, C. and Möller, P., Nucl. Phys. **A115**, 545 (1968).
22. Ragnarsson, I. and Nilsson, S. G., Progress report on experimental and theoretical studies, Colloquium on intermediate nuclei June 30–July 2, 1971, Orsay.
23. Strutinsky, V. M., Nucl. Phys. **A95**, 420 (1967).
24. Strutinsky, V. M., Nucl. Phys. **A212**, 1 (1968).
25. Bolsterli, M., Fiset, E. A., Nix, J. R. and Norton, J. L., Phys. Rev. C, to be published.
26. Brack, M., Damgaard, J., Pauli, H. C., Stenholm-Jensen, A., Strutinsky, V. M. and Wong, C. Y., Rev. Mod. Phys. (1972), to be published.
27. Cline, D., to be published in the Proceedings of the Orsay Colloquium on Intermediate Nuclei, 1971.
28. Gustafson, C., Lamm, I. L., Nilsson, B. and Nilsson, S. G., Ark. Fys. **36**, 613 (1967).
29. Larsson, S. E., Bjørnholm, S. and Nilsson, S. G., to be published.
30. Bjørnholm, S. and Lynn, L., Rev. Mod. Phys., to be published.
31. Pauli, H. C. and Ledergerber, T., Nucl. Phys. **A175**, 545 (1971).
32. Bjørnholm, S., Dubois, J. and Elbek, B., Nucl. Phys. **A118**, 241 (1968)
33. Tsang, C. F. and Nilsson, S. G., Nucl. Phys. **A140**, 289 (1970).

Department of Mathematical Physics
Lund Institute of Technology
P.O. Box 725
S-220 07 Lund 7, Sweden